Image Processing for Mechatronics MCTR1010
Mechatronics Engineering
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Goalkeeper Robotic Arm

By

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Chapter 1

Literature Review

Vision-based robotic systems have become an essential research area in mechatronics and robotics, particularly for applications that require interaction with dynamic environments. One of the key applications is robotic goalkeeping, where the robot must detect a moving ball, estimate its trajectory, and respond rapidly to intercept it. Image processing techniques combined with control algorithms allow robots to perform such tasks with increasing accuracy and speed. Several studies have explored different approaches to ball detection, trajectory prediction, and robotic actuation using camera-based sensing systems.

Ball detection and tracking is the first step in implementing a vision-based goalkeeper system. Kim et al. [1] proposed a real-time ball tracking system using classical computer vision techniques implemented in OpenCV. Their method relies on color segmentation to isolate the ball from the background by converting the captured image into the HSV color space. After segmentation, contour detection and centroid extraction are used to determine the exact position of the ball in each frame. The system was able to track a moving ball efficiently in real time, making it suitable for robotic applications where fast reaction times are required. The study demonstrated that lightweight image processing techniques can be effectively implemented on embedded platforms such as Raspberry Pi for real-time object tracking.

In addition to detecting the ball position, predicting its future trajectory is critical for robotic goalkeeper systems. Liu et al. [2] introduced a vision-based trajectory prediction approach using a Kalman filtering algorithm. In this method, the ball's position and velocity are estimated from consecutive camera frames, and a predictive model is used to estimate the future path of the ball. By predicting the trajectory in advance, the robot can move proactively rather than reacting only after the ball reaches a certain location. The results of the study showed that trajectory prediction significantly improves interception accuracy, particularly when dealing with high-speed moving objects such as soccer balls.

Recent advances in deep learning have also improved object detection performance in robotic

vision systems. Bochkovskiy et al. [3] introduced the YOLOv4 object detection framework, which provides a balance between detection accuracy and computational efficiency. YOLO-based detection models are capable of identifying objects within images in real time, making them suitable for robotic perception tasks. For embedded systems with limited computational power, lightweight versions such as YOLOv4-Tiny can be deployed to detect objects such as balls or obstacles with relatively high speed. These models allow robotic systems to operate in more complex environments where traditional color-based detection methods may struggle due to variations in lighting conditions or background clutter.

Another approach for detecting objects in robotic systems is color-based segmentation combined with morphological operations. Sharma et al. [4] proposed a color-based object tracking method designed for mobile robots. In their work, images captured by a camera are converted from RGB to HSV color space to isolate objects with specific color ranges. The segmented regions are then processed using morphological operations such as erosion and dilation to remove noise and improve detection accuracy. Finally, contour analysis is used to determine the centroid of the detected object, which can then be used for robot navigation or object tracking tasks. This approach is computationally efficient and well suited for systems that require real-time processing using low-cost hardware platforms.

A specific application closely related to the current project is the implementation of robotic soccer goalkeepers. Al-Khalifa et al. [5] developed a vision-based robotic goalkeeper capable of detecting and intercepting a soccer ball using a camera and a servo-actuated arm. In their system, the camera continuously captures frames of the playing field, and the ball is detected using image segmentation techniques. The trajectory of the ball is estimated based on its motion between frames, allowing the system to predict where the ball will cross the goal line. A control algorithm then commands the robotic arm to move to the predicted interception point. Experimental results demonstrated that the system was capable of successfully blocking a significant percentage of shots, highlighting the potential of integrating computer vision with robotic control systems.

In general, most vision-based robotic systems follow a similar processing pipeline. First, images are acquired from a camera sensor connected to the processing unit, which may be a laptop, embedded computer, or microprocessor such as a Raspberry Pi. Next, image preprocessing techniques such as filtering and noise reduction are applied to enhance the quality of the captured images. Feature extraction methods are then used to identify relevant objects within the scene, such as balls, obstacles, or markers. Finally, control algorithms process the extracted features to generate commands for actuators such as motors or servos. This architecture allows robotic systems to convert visual information into physical actions, enabling interaction with the environment.

For embedded robotic systems, computational efficiency is an important design consideration. Platforms such as Raspberry Pi provide sufficient computational power for many com-

puter vision tasks while maintaining low cost and energy consumption. When combined with libraries such as OpenCV, these platforms can perform real-time image acquisition, filtering, segmentation, and object tracking. In addition, the Raspberry Pi can communicate with microcontrollers such as Arduino to control motors and actuators. This distributed architecture allows the Raspberry Pi to perform high-level decision making based on image processing results while the Arduino handles low-level motor control tasks.

Overall, the literature demonstrates that vision-based robotic goalkeeping systems rely heavily on robust object detection and trajectory prediction algorithms. Techniques such as color segmentation, contour detection, Kalman filtering, and deep learning-based object detection have been successfully applied in various robotic applications. By combining these approaches with embedded processing platforms and simple control mechanisms, it is possible to design an effective robotic goalkeeper capable of detecting, predicting, and intercepting a moving ball in real time.

Chapter 2

Hardware Design

2.1 Component List

Table 2.1: Hardware Components Required for the Goalkeeper Robotic Arm

Component Name	**Quantity**	**Approx. Price (EGP)**	**Purchase Source**
Raspberry Pi 4 Model B (4GB)	1	4750	RAM Electronics
Raspberry Pi Camera Module V2	1	800	RAM Electronics
Raspberry Pi Camera Cable	1	185	RAM Electronics
Arduino Uno R3	1	750	RAM Electronics
MicroSD Card 64GB	1	495	RAM Electronics
High Torque Servo Motor (FS5103B)	1	300	RAM Electronics
USB Cable (Arduino Connection)	1	40	RAM Electronics

2.2 Project Plan

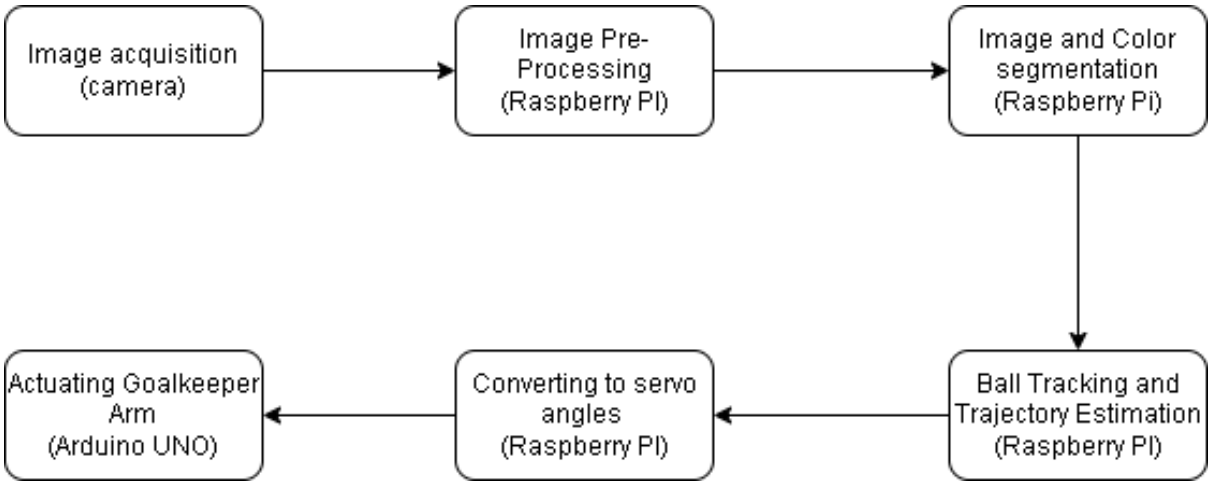

Figure 2.1: Project Plan

2.3 Solidworks Design

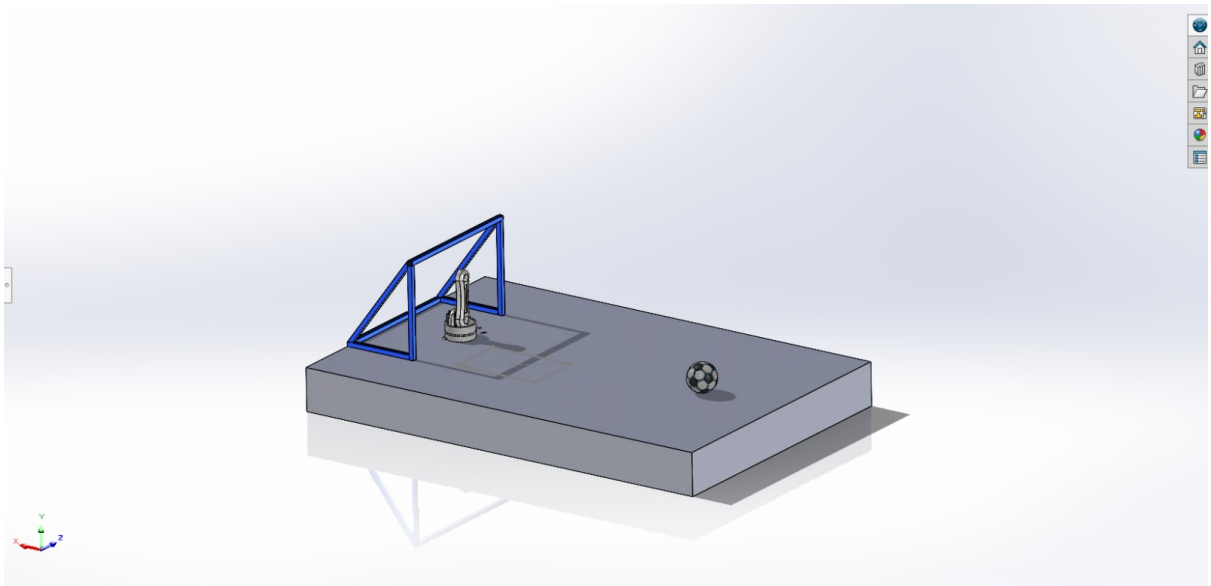

Figure 2.2: SolidWorks

Chapter 3

Circuit Design

For the circuits design the Raspberry Pi 4b will be the main controller with Camera connected to iy. And the Arduino UNO will be connected to RPI through the USB Cable. Finally, Servo motor will be connected to the Arduino UNO.

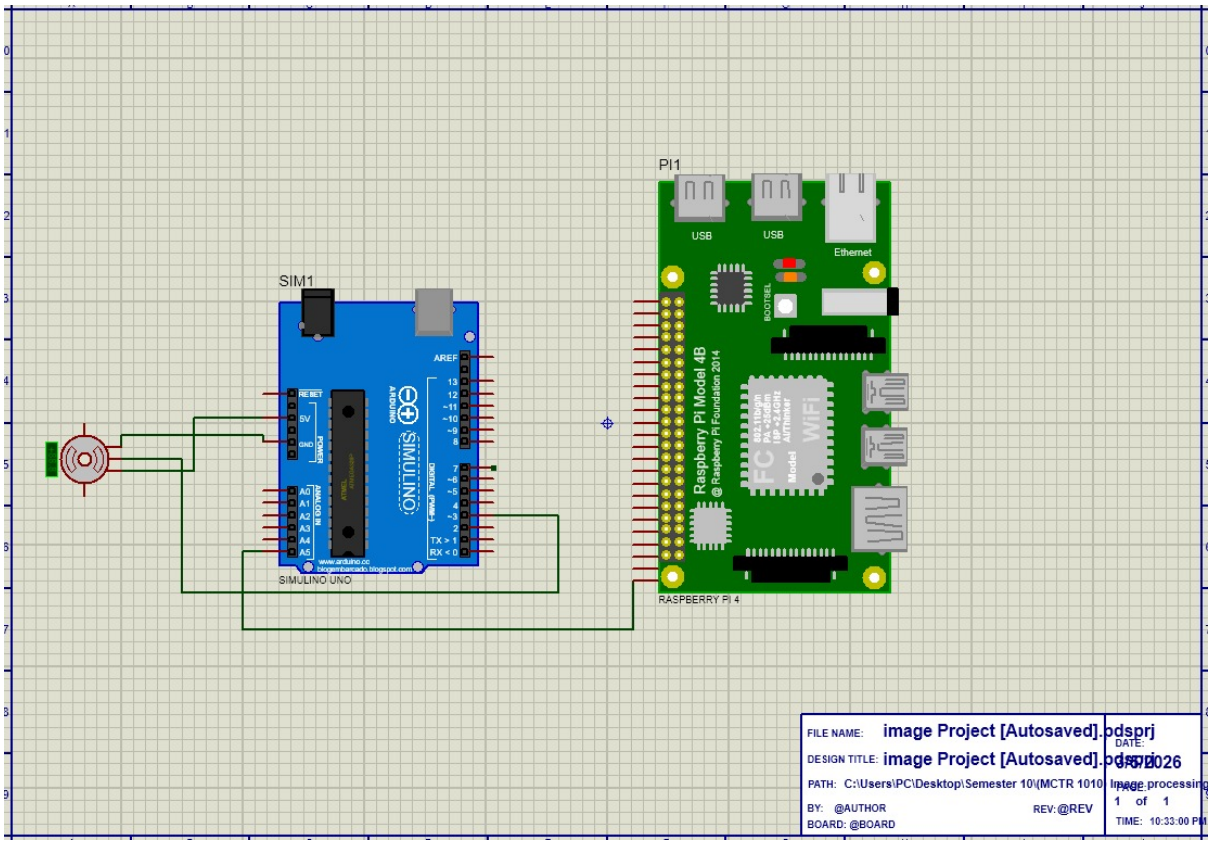

Figure 3.1: Circuits Design

Chapter 4

Raspberry Pi Capture

Figure 4.1: Image Captured by RPI Camera

Bibliography

[1] S. Kim, J. Lee, and H. Park. Real-time ball tracking using computer vision for robotic applications. *IEEE Access*, 9:11234–11245, 2021.

[2] J. Liu, T. Wang, and Y. Zhang. Vision-based ball trajectory prediction for robotic goalkeepers. *Robotics and Autonomous Systems*, 150:103987, 2022.

[3] A. Bochkovskiy, C. Wang, and H. M. Liao. Yolov4: Optimal speed and accuracy of object detection. *Computer Vision and Pattern Recognition*, 2021.

[4] P. Sharma, R. Gupta, and K. Singh. Color-based object detection and tracking using opencv for mobile robots. *Sensors*, 23(8):4125, 2023.

[5] M. Al-Khalifa, M. Rahman, and S. Ahmed. Design and implementation of a vision-based soccer goalkeeper robot. *IEEE Access*, 10:34567–34578, 2022.